

NETWORKS AND PROTOCOLS

Lab 01

Objectives

Get you familiar with basic networking tools.

Exercise 01: `ifconfig` (Network interfaces and their names)

The `ifconfig` utility lists a computer's network interfaces and displays or updates their configuration. Type `ifconfig` in the command line and answer the following questions:

Note that the equivalent command in Windows is `ipconfig`.

- (a) How many network interfaces does your computer have?
- (b) Can you guess why it has more than one?
- (c) What is the IP address of each interface?
- (d) What is the MAC address of each interface?
- (e) Why could it be that some interfaces do not have a MAC address?

Exercise 02: Use `ping` to test host reachability

Are the following hosts reachable from your machine by using `ping`:

- `www.ensia.edu.dz`
- `www.algeria.dz`
- `www.kremlin.ru`
- `www.cerist.com`
- `www.hola.hp`
- `8.8.8.8`
- `www.mit.edu`
- `www.amazon.com`
- `www.intel.com`
- `www.tsinghua.edu.cn`

If you observe that some hosts are not reachable, then can you explain why? Check if the addresses that were unreachable by the `ping` command are reachable from the Web browser (here you should use the website's URL instead of the IP address).

Here are some of `ping` options for Windows :

- `-t`: Sends packets continuously until the user interrupts.

- -n count: Sends count packets, then stops.
- -l packetsize: Specifies packet size in bytes.
- -w timeout: Sets the timeout in seconds for each request

Here are some of **ping** options for Unix-like operating systems:

- -c count: Sends count packets, then stops.
- -s packetsize: Specifies packet size in bytes.
- -i timeout: Sets the timeout in seconds for each request
- -o: Exit successfully after receiving one reply packet.
- -a Include a bell character in the output when any packet is received.

Exercise 03: Use **traceroute** to understand network topology

When two computers (end-systems) communicate with each other over the Internet, their communication traverses multiple packet switches. There are two general types of packet switches on the Internet: link-layer switches and network-layer switches (the latter are also called routers).

The **traceroute** utility lists the routers that are located between your computer and a remote one. E.g., if you type **traceroute <target>** in the command line, where **<target>** is a website name or IP address, it will display a list of router names and/or IP addresses and the RTTs (Round Trip Times) that were measured between your computer and each router.

- Run **traceroute** on your machine to **www.mit.edu**
- How many routers are there between your workstation and **www.mit.edu** ?
- Between which two routers do packets cross the Atlantic Ocean?

Run **traceroute** from your machine to the following destinations: (I)**www.ucla.edu** (II)**www.utokyo.ac.jp** and (III)**www.lancaster.ac.uk**.

- At which router do the paths from your machine to these three destinations diverge?
- Find out further details about this router.
- Is the number of hops on each path proportional to the physical distance?

Several servers distributed around the world provide a web interface from which you can perform a **traceroute** to any other host in the Internet. Here are two examples:

(I) <http://www.speedtest.com.sg/tr.php> and (ii) <https://www.telstra.net/cgi-bin/trace>. You can find other traceroute servers at www.traceroute.org

- (a) Run `tracert` from both these servers towards your machine and in the reverse direction (i.e. from your machine to these servers)
- (b) What are the IP addresses of the two servers that you have chosen?
- (c) Does the reverse path go through the same routers as the forward path?
- (d) If you observe common routers between the forward and the reverse path, do you also observe the same IP addresses? Why or why not?

The equivalent command in Windows is `tracert`.

Here is some of `tracert` options in Windows:

- -d: Do not resolve addresses to hostnames.
- -h `maxiHops`: Maximum number of hops to search for target.
- You can introduce the packet size in `tracert` command, exp: `tracert 1.1.1.1 133`

Here is some of `tracert` options in Unix-like operating systems:

- -m `MaxHop`: Sets the maximum number of hops used in outgoing probe packets.
- -n: Do not resolve addresses to hostnames.
- -q `Nqueries`: Specifies the number of probes the `tracert` command sends at each hop. The default is three probes.
- -v: Receives packets other than `TIME_EXCEEDED`.
- -w `WaitTime`: Sets the time (in seconds) to wait for a response to a probe. The default is 3 seconds.

Exercise 04: Use ping to gain insights into network performance

For each of these locations (`www.qu.edu.qa` , `www.nd.edu` , `www.imperial.ac.uk`) find the (approximate) physical distance from ENSIA using Google Maps and compute the shortest possible time T for a packet to reach that location from ENSIA. You should assume that the packet moves (i.e. propagates) at the speed of light, 3×10^8 m/s. Note that the shortest possible time will simply be the distance divided by the propagation speed. Plot a graph where the x-axis represents the distance to each city (i.e. Doha, Indiana and London), and the y-axis represents the ratio between the minimum delay (i.e. RTT) as measured by the ping program (select the values for 50 byte packets) and the shortest possible time T to reach that city from ENSIA. (Note that the y-values are no smaller than 2 since it takes at least $2 \times T$ time for any packet to reach the destination from ENSIA and get back).

- (a) Can you think of at least two reasons why the y-axis values that you plot are greater than 2?

- (b) Is the delay to the destinations constant or does it vary over time? Explain why.
- (c) The measured delay (i.e., the delay you can see in the graphs) is composed of propagation delay, transmission delay, processing delay and queuing delay. Which of these delays depend on the packet size and which do not?

Exercise 05: dig

Humans use special names to refer to computers (more precisely, to the network interfaces of computers). When you instruct your computer to communicate with a remote computer that has a given name, your computer translates the given name to an IP address.

- What are the IP addresses of `www.ensia.edu.dz`?
- What are the IP addresses of `www.yahoo.fr` ?
 - Why could it be that `www.yahoo.fr` maps to more than one IP addresses?
- What is the IP address of `www.google.com`?
- Answer the same question again in an hour or so. Has anything changed? If so,
 - What could be the reason for the change?
 - Which is the IP address of the Google site (`www.google.com`)?
 - In your opinion, what is the reason of having several IP addresses as an output?
- Find out name of the IP address `127.0.0.1`. What is special about this IP address

The equivalent command in windows is `nslookup`.

Exercise 06: netstat

The `netstat` command is used to display the TCP/IP network protocol statistics and information. To use it, type `netstat` in the command line.

- What are the opened ports on your computer ?
- What are the remote addresses (or names) that communicate with your computer ?

Here are some `netstat` options:

- `-a`: Show the state of all sockets.
- `-b`: Show the number of bytes in and out.
- `-m`: Show statistics recorded by the memory management routines.
- `-n`: Show network addresses as numbers.
- `-p protocol`: Show statistics about protocol (ex: `tcp/udp`).
- `-s`: Show per-protocol statistics.